C# & .NET Basic Certification Exam — Paper A

Focus: .NET Environment, Visual Studio, Variables, Operators & Control Flow

Course: Introduction to C# Programming	**Duration:** 60 Minutes
Topic: Basic .NET & C# Fundamentals	**Total Marks:** 50 Marks
Instructions: Write all code legibly. Answer all questions in Sections A, B, and C.	**Passing Grade:** 60%

Section A: .NET Environment & Core C# Syntax (15 Marks)

1. (3 Marks) Which component of the .NET Framework is responsible for executing C# applications, managing memory, and providing garbage collection?

A) Visual Studio Community IDE
B) Common Language Runtime (CLR)
C) Just-In-Time (JIT) Compiler only
D) C# Source Code Editor

2. (3 Marks) What is the function of the Just-In-Time (JIT) compiler in the .NET Framework execution pipeline?

A) Converts C# source code directly into HTML and CSS web files.
B) Compiles C# source code into C++ native code before installation.
C) Converts Intermediate Language (IL) code into native machine code at runtime.
D) Checks C# code for syntax errors inside Microsoft Visual Studio.

3. (4 Marks) Consider the variable declarations below. Which line contains a compile-time error, and why?

```
int age = 25;
double rating = 4.8;
int total = rating; // Line 3
```

Explain how to fix Line 3 using explicit casting in C#.

4. (5 Marks) Briefly describe the role of Microsoft Visual Studio Community in C# application development. List two key features provided by the IDE.

Answer lines:

Section B: Conditional Statements & Logical Operators (15 Marks)

5. (5 Marks) What will be printed to the Console when the following C# snippet executes?

```
int x = 10;
int y = 20;
if (x > 5 && y < 15) {
    Console.WriteLine("Option A");
} else if (x == 10 || y == 10) {
    Console.WriteLine("Option B");
} else {
    Console.WriteLine("Option C");
}
```

6. (5 Marks) Write a short C# statement to read user input from the console and convert it safely into an integer named 'userNumber'.

Write your C# code below:

7. (5 Marks) Compare a 'switch' statement with an 'if-else if' ladder in C#. State one scenario where using a 'switch' statement is preferred.

Section C: Loops & Practical Problem Solving (20 Marks)

8. (8 Marks) Determine the exact console output of the following loop:

```
for (int i = 1; i <= 5; i++) {
    if (i == 3) continue;
    Console.Write(i + " ");
}
```

9. (12 Marks) Coding Task: Write a complete C# console snippet that uses a 'while' loop to repeatedly ask the user to enter integers until they enter 0. The program should then print the sum of all entered numbers.

Write your C# solution below:

Exam Paper A — Solutions & Marking Scheme

Question 1 Answer: B

The Common Language Runtime (CLR) is the core execution engine of .NET that manages application execution, memory allocation, and garbage collection.

Question 2 Answer: C

The JIT compiler converts Intermediate Language (IL) bytecode into native CPU instructions right before execution on the host machine.

Question 3 Answer: Line 3 fails compile-time check.

Cannot implicitly convert type 'double' to 'int'. Fix: int total = (int)rating; (explicit narrowing conversion).

Question 4 Answer: Visual Studio IDE

Visual Studio Community is a free, extensible IDE for building .NET apps. Features: Code IntelliSense/Auto-completion, integrated Debugger, and Solution Explorer.

Question 5 Answer: Option B

Explanation: (x > 5 && y < 15) is (True && False) = False. Next condition: (x == 10 || y == 10) is (True || False) = True. Thus, 'Option B' prints.

Question 6 Answer: Console Parsing Code

string input = Console.ReadLine(); int userNumber = int.Parse(input); // or int.TryParse(input, out userNumber);

Question 7 Answer: Switch vs If-Else

A switch statement tests a single variable against discrete constant values. It is cleaner and more readable when checking multiple fixed values (e.g. enum or menu choices) compared to repetitive if-else statements.

Question 8 Answer: Output is '1 2 4 5 '

The continue statement skips iteration when i == 3, so 3 is not printed.

Question 9 Answer: While Loop Summing Code

int sum = 0; int number = -1; while (number != 0) { Console.Write("Enter a number (0 to stop): "); number = int.Parse(Console.ReadLine()); sum += number; } Console.WriteLine("Total sum: " + sum);